

SolarGen Production Forecast Model

SolarGen

Model version: hourly-capped production blend selected through 2026-09-25

1 Scope

SolarGen forecasts rooftop photovoltaic generation for a 10 kWp south-facing rooftop array in OHZ / Osterholz-Scharmbeck. The production forecast shown in the browser and local history app is an equal hourly blend of an Open-Meteo current physical forecast and a DWD stable forecast, capped at the observed 6.1 kWh hourly curtailment level.

The browser app obtains DWD data through the Open-Meteo DWD ICON endpoint. The local history app obtains DWD data from DWD MOSMIX. Both surfaces use the same transfer structure and final production blend.

2 Source Physical Model

For each source $s \in \{\text{OM}, \text{DWD}\}$, hourly meteorological inputs are:

$$I_{s,h}, C_{s,h}, R_{s,h}, T_{s,h},$$

where I is global tilted irradiance in W/m^2 , C is cloud cover, R is precipitation, and T is ambient temperature.

The source physical model computes a rooftop generation curve:

$$G_{s,h}^{\text{cur}} = f_{\text{site}}(I_{s,h}, C_{s,h}, R_{s,h}, T_{s,h}).$$

The daily current total for source s is:

$$G_s^{\text{cur}} = \sum_{h=0}^{23} G_{s,h}^{\text{cur}}.$$

3 DWD Stable Transfer

The DWD current physical model alone did not generalize well. SolarGen therefore transforms DWD through a stable daily transfer model before production blending.

First compute a DWD sunshine/rain daily model:

$$G_{\text{DWD}}^{\text{raw}} = \max(0, 18.3545 + 2.3510H_{\text{DWD}} - 1.9219R_{\text{DWD}}^+),$$

where H_{DWD} is DWD sunshine duration in hours and R_{DWD}^+ is DWD precipitation summed only over hours with positive tilted irradiance:

$$R_{\text{DWD}}^+ = \sum_{h=0}^{23} \mathbf{1}\{I_{\text{DWD},h} > 0\} R_{\text{DWD},h}.$$

The stable DWD daily forecast is:

$$G_{\text{DWD}}^{\text{stable}} = 0.25G_{\text{DWD}}^{\text{cur}} + 0.75G_{\text{DWD}}^{\text{raw}} + 4.039$$

4 Production Blend

There is no production-bias term or fitted production coefficient. The uncapped hourly blend is defined below, then capped at 6.1 kWh to obtain forecastable production. Uncapped output is retained as theoretical potential.

5 Hourly Allocation

The DWD current hourly curve is scaled to the stable DWD daily total:

$$\tilde{G}_{\text{DWD},h} = G_{\text{DWD},h}^{\text{cur}} \cdot \frac{G_{\text{DWD}}^{\text{stable}}}{\sum_{j=0}^{23} G_{\text{DWD},j}^{\text{cur}}}.$$

The uncapped production hourly curve is:

$$G_h^{\text{uncap}} = 0.50G_{\text{OM},h}^{\text{cur}} + 0.50\tilde{G}_{\text{DWD},h}.$$

The curve is scaled by a small numerical factor if needed so that its uncapped total equals the daily source blend. Forecastable hourly and daily production are:

$$G_h^{\text{prod}} = \min(6.1, G_h^{\text{uncap}}), \quad G^{\text{prod}} = \sum_{h=0}^{23} G_h^{\text{prod}}.$$

6 Curtailment, Load, Battery, And Value

The browser app applies the household load, battery, export, and value model to G_h^{prod} . For each hour:

$$G_h^{\text{deliv}} = \min(G_h^{\text{prod}}, F),$$

$$G_h^{\text{curt}} = G_h^{\text{uncap}} - G_h^{\text{deliv}},$$

where $F = 6$ kW by default. Delivered PV first covers same-hour household load, then charges the battery, and remaining surplus is exported.

7 Selection Evidence

The production model was selected on all 129 paired Open-Meteo and DWD day-ahead dates with actuals from 2026-05-14 through 2026-09-25. The validation target caps hourly actual generation at 6.1 kWh so above-curtailment production from coincident household consumption does not drive the fit.

Model	In-sample MAE	LOO MAE	Rolling MAE	Comment
Huber capped weather	4.910	5.243	5.736	12 params
Hourly-capped equal blend	5.313	5.313	5.736	selected
Linear OM+DWD+bias	5.297	5.418	5.748	3 params
Previous uncapped equal	5.455	5.455	5.867	previous
Old 0.73 blend	5.541	5.541	5.985	older

The selected model is not the best in-sample or leave-one-out model. Huber regression improves leave-one-out MAE by only 0.070 kWh/day and ties the selected model on chronological rolling MAE while using 12 fitted degrees. The selection rule chooses the fewest fitted parameters within one standard error of the best rolling MAE.

Relative to the previous uncapped equal blend, the selected model improves aggregate leave-one-out MAE from 5.455 to 5.313 kWh and rolling RMSE from 8.529 to 8.025 kWh. The paired bootstrap 95 percent interval for MAE improvement includes zero, and June regresses by 0.504 kWh/day; the gain is therefore modest rather than decisive.

8 Current Stored-History Accuracy

After recomputing production rows under the hourly-capped equal blend, the local history database contains 129 paired actual days:

Metric	Value
MAE	5.313 kWh
RMSE	7.502 kWh
Bias	-0.631 kWh
WAPE	14.982 percent

9 Implementation References

- `src/model.js`: source physical model.
- `src/productionBlend.js`: browser production blend.
- `src/weather.js`: Open-Meteo and DWD ICON browser endpoints.
- `history_app/forecast_model.py`: local production blend using stored OM/DWD inputs.
- `history_app/cli.py`: `recompute-production` command.

10 Maintenance Rule

Whenever model parameters change, update this document, regenerate the PDF, recompute production history with:

```
python3 -m history_app.cli recompute-production
```

and run:

```
npm run check    npm test
```